

Distributions in Data Science – Notes

1. What is Distribution?

- Distribution in data science is a method which shows the probable values for a variable and how often they occur. It is a simple way to visualize a set of data..
- It shows:
 - Possible values of a variable
 - How often each value occurs (probabilities)

Example:

- Coin toss:
 - Head = 0.5
 - Tail = 0.5
 - Any other outcome = 0

Important Points

- Distribution includes **all possible outcomes**.
- It is often shown using **graphs**, but:
 - Graph is only a **visual representation**
 - Actual distribution is defined by **probabilities**

Example of Distribution

Tossing Two Coins

Possible outcomes:

HH, HT, TH, TT

Probability:

Each = 0.25

This is an example of Uniform Distribution
(all outcomes equally likely)

Types of Data in Distribution

The data can be classified into two types

1. Discrete Data
2. Continuous Data

(A) Discrete Data

- Takes **fixed values only**
- Example:
 - Pass / Fail
 - Number of students

(B) Continuous Data

- Can take **any value in a range**
- Example:
 - Height
 - Weight
 - Temperature

So based on type of data, distribution can be of two types.

Discrete probability distribution

Continuous probability distribution

Poisson Distribution

- Used to find probability of events happening in a fixed interval
- Example: Number of calls received in a call center per hour

Binomial Distribution

- Used when there are only two outcomes (Success/Failure)
- Example: Tossing a coin multiple times

Bernoulli distribution

It is the **simplest probability distribution** in data science.

It deals with **only one trial** having **two possible outcomes**:

- Success (1)
- Failure (0)

Examples:

- Toss a coin → Head (1), Tail (0)
- Exam result → Pass (1), Fail (0)
- Light bulb → Working (1), Not working (0)